MIL-C-00401F(AR)
18 JULY 1986
USED IN LIEU OF
MIL-C-401E
13 March 1974

MILITARY SPECIFICATION

COMPOSITION B

This Military Specification has been prepared by the US Army Armament Munitions and Chemical Command based upon currently available technical information but has not been approved for promulgation as a coordinated revision of MIL-C-401E. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provision and packaging for the manufacture of two types of high explosives designated as Composition B intended for use as main explosive charges in artillery ammunition.

1.2 Classification. Composition B shall be of the following types and grades as specified (see 6.1):

TYPE I - Flake with Desensitizing Wax

GRADE A	Having an efflux viscosity value of 7.0 sec (max)
GRADE B	Having an efflux viscosity value of 17.0 sec (max)

TYPE II - Granular with Desensitizing Wax

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 1376
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:21Z
Check the source to verify that this is the current version before use.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-R-398	- RDX
MIL-T-248	- Trinitrotoluene (TNT)
MIL-W-20553	- Wax, Desensitizing
MIL-I-45208	- Inspection System Requirements

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-650	- Explosives: Sampling, Inspection and Testing
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1171	- Acceptance and Description Sheets (For Propellants and Explosives)
MIL-STD-1218	- ACS Chemicals
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

7548644	- Box, Packing for High Explosives, Assembly, Details, Packing and Marking
7548645	- Box, Fiberboard, Packing, Reusable, Collapsible for High Explosives.
9257923	- Box, Fiberboard, Packing, Reusable, Collapsible, for High Explosives, Single Piece.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:21Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings and publications, required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

CODE OF FEDERAL REGULATIONS

49 CFR 100-100 - Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations, available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publications should cite "49 CFR".)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Designation E300 - Recommended Practice for Sampling Industrial Chemicals

ASTM D2905 - Statement on Number of Specimens Required to Determine the Average Quality of a Textile Material

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:21Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material.

3.1.1 Composition B, types I and II. The composition shall contain RDX complying with the requirements of MIL-R-398, Desensitizing Wax complying with the requirements of MIL-W-20553 and Type I TNT complying with requirements of MIL-T-248. These materials shall be uniformly incorporated in the proportions specified in 3.2 to form a homogeneous mixture free from contamination.

3.2 Physical and chemical properties. Composition B shall conform to the requirements specified in Table I, when tested as specified in the applicable test methods.

TABLE I. Properties and requirements

Property	Type I	Type II	Applicable Test Method Paragraph
RDX and HMX %	59.5 ± 2.0	59.5 ± 2.0	4.6.1 or 4.6.2
HMX, % max	14.90		
TNT, %	39.5 ± 2.3	39.5 ± 2.3	4.6.1 or 4.6.4
Wax %	1.0 ± 0.3	1.0 ± 0.3	4.6.3
Moisture, % max	0.20	0.20	4.6.5
Bulk Density, gm/cc			4.6.6
min.	-	0.90	
Insoluble Particles,			4.6.7
Retained on USS No. 60	5	0	
Retained on USS sieve No 40		0	
Viscosity, Efflux			4.6.8
sec., max.	-	-	
Grade A	7.0	-	
Grade B	17.0	-	
Granulation, (%)			4.6.9
Thru USS No. 10 Screen, min.	-	100	
Thru USS No. 200 Screen, max.	-	1.0	
Workmanship	-	-	4.6.10

3.3 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:21Z
Check the source to verify that this is the current version before use.

3.4 Process controls. The contractor shall submit a Process Control Document to the Government specifying the process variables which are considered critical for the production of Composition B. The Process Control Document shall be submitted in accordance with 4.2.

3.5 Workmanship. The manufacturer shall use procedures and controls which assure that the Composition B produced is uniformly incorporated to form a homogeneous composition and does not contain foreign material such as dirt, rust, paint or metal chips, etc., and that the safety and reliability of the explosive are not compromised. Compliance with this requirement shall be as specified in 4.6.10.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspections specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspections specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements. Reference shall be made to MIL-STD-109 to definne terms used herein.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shalll become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Process control data. A Process Control Document shall be submitted to the Technical Agency (AMSMC-QAR-R (D)) in accordance with Data Item Description, DI-P-10392. The document shall contain a description of the process, all materials used, process conditions/procedures and production/inspection equipment used to produce Composition B meeting the requirements of this specification (see 6.1.2). In addition, whenever there is a change (change in source of materials, process conditions, procedures, etc.), from previously reported process control data, the Technical Agency shall be notified of the new process control data and the reason for the change.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:21Z
Check the source to verify that this is the current version before use.

4.3 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

4.4 First article inspection.

4.4.1 Submission. When specified in the contract or purchase order, the contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.4.2. The first article sample shall be ten (10) pounds of Composition B. The sample shall be obtained from the first production lot which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

4.4.2 Inspections to be performed. The sample may be subjected by the Government to any or all of the examinations and tests specified in 4.5 and 4.6 of this specification and any or all requirements of the applicable drawings.

4.4.3 Rejection. If any sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements.

4.5 Quality conformance inspection.

4.5.1 Lot formation. A lot shall consist of one or more batches of Composition B produced by one manufacturer, in accordance with the same specification or specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of Composition B that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. For continuously produced Composition B, a unit (batch) is one hour of continuous production (production interval). In addition, each lot shall contain:

a. TNT from one interfix lot only, from one manufacturer.

b. RDX from one interfix lot only, from one manufacturer.

c. Densensitizing wax from one lot only from one manufacturer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:21Z
Check the source to verify that this is the current version before use.

The product shall be submitted for inspection in accordance with MIL-STD-105 unless otherwise specified. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.5.2 Examination. Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105. Contractor's sampling plans, if used, shall be approved by the Government and shall provide as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:21Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-00401F

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.5.2.1	Wooden or fiberboard box				7548644/7548645
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Foreign matter		1.0		Visual
102	Liner pierced or torn		1.0		Visual
103	Liner improperly closed		1.0		Visual
Minor					
201	Type of liner incorrect		1.0		Visual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:21Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-00401F

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7548644
4.5.2.2	Sealed wooden box				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box damaged		1.5		Visual
102	Lot number incorrect or illegible		1.5		Visual
103	Board broken or split		1.5		Visual
104	Strapping missing broken or loose		1.5		Visual/manual
105	Top improperly assembled		1.5		Visual/manual
Minor					
201	Nail protruding		1.5		Visual
202	Marking incorrect or illegible		1.5		Visual
203	Strapping improperly assembled		1.5		Visual/manual

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:21Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-00401F

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.5.2.3	Sealed fiberboard box				7548645
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Assembly torn or pierced		1.0		Visual
102	Lot number incorrect or illegible		1.0		Visual
103	Binding strips broken, missing or loose		1.0		Visual/manual
Minor					
201	Marking incorrect or illegible		1.5		Visual
202	Stitches missing or loose (when required)		1.5		Visual/manual
203	Banding strips improperly assembled		1.5		Visual/manual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:21Z
Check the source to verify that this is the current version before use.

4.5.3 Testing.

PRECAUTION WARNING

This specification covers sampling and testing of toxic or hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.3.1 Sampling.

4.5.3.1.1 Sampling of composition B . The tests in 4.6 shall be performed on samples representative of the batch which were taken in accordance with ASTM Procedure E300, for solids. Approximately 1 kg sample shall be obtained from each batch. The selection of batches for testing shall be in accordance with MIL-STD-1235, CSP-1 Plan, AQL 6.5%. A batch of Composition B produced by a batch method shall be as specified in 4.5.1. For continuously produced Composition B a batch for sampling purposes shall be that quantity produced during a discrete one hour duration of continuous production. If any sample fails to meet any test requirement, the batch represented by the sample shall be rejected. All batches produced between the time that the last batch was tested and accepted and the batch which failed shall be tested in accordance with the applicable methods given in paragraph 4.6, unless otherwise directed by the contracting officer (see 6.3). If any of these batches fail to meet any of the test requirements, that batch shall also, be rejected. In addition, after any failure of a batch the contractor will return to 100% inspection until "i" successive batches are accepted in accordance with MIL-STD-1235.

4.5.3.2 Product testing. Unless otherwise specified, the samples obtained in 4.5.3.1 shall be subjected to the tests listed in Table II. The viscosity determination shall be performed on every batch of Type I Composition B. Unless otherwise specified in the contract (see 6.1) the results of the tests shall be reported in accordance with MIL-STD-1171.

4.5.3.3 Classification of defects. The classification of defects shall be as given in Table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:21Z
Check the source to verify that this is the current version before use.

TABLE II. Required tests

Inspection/Requirement (See 3.2, Table 1)	Defect Classification	Test Method
RDX, %	Major	4.6.1 or 4.6.2
Wax, %	"	4.6.3
TNT, %	"	4.6.1 or 4.6.4
Moisture, %	"	4.6.5
Bulk Density	"	4.6.6
Insoluble Particles	"	4.6.7
Viscosity	"	4.6.8
Granulation	"	4.6.9
Workmanship	"	4.6.10

4.5.4 Inspection equipment. For the performance of all tests and examinations specified in 4.5 and 4.6, commercial inspection equipment should be employed. The contractor shall have available, and utilize correctly, this equipment and is charged with the responsibility of assuring that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes (see 6.4).

4.6 Methods of inspection. The following tests shall be performed using prescribed analytical procedures for replicate determinations given in standard analytical textbooks or ASTM Method D2905. Also, unless otherwise specified herein, all chemicals and reagents shall be ACS grade in accordance with MIL-STD-1218 or reagent grade (see 6.2 for use of equivalent test methods).

4.6.1 RDX, HMX and TNT content.

4.6.1.1 Sample preparation. Crush and screen a portion of the sample. Collect that portion of the material passing through a USSS No. 4 for the test sample used in this analysis and the subsequent analyses in 4.6.

4.6.1.2 Preparation of sample solution. Accurately weigh a 1.0g $\pm$ 0.0050g portion of the sample obtained in 4.6.1.1 to the nearest 0.0001g into a tared 50 ml beaker. With the aid of a stream of UV Grade acetonitrile from a wash bottle, quantitatively transfer the weighed sample into a 100 ml volumetric flask. Add additional solvent to the flask for a total volume of approximately 80 ml. Place the flask in an ultrasonic bath until the sample is completely dissolved in the solvent. Remove the flask from the bath, allow the flask and contents to reach equilibrium at room temperature. Fill the flask with the solvent to the 100 ml mark and mix thoroughly. Use this solution for the RDX/TNT determination.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:21Z
Check the source to verify that this is the current version before use.

4.6.1.3 Calibration. Establish calibration factors by analyzing known concentrations of RDX, HMX and TNT using the High Performance Liquid Chromatographic (HPLC) method. Use various concentrations of RDX, HMX and TNT covering the maximum and minimum concentrations given in this specification.

4.6.1.4 Standard sample. Prepare and analyze a standard sample each day the instrument is used to determine acceptability of material. Use the procedures of analysis given in this paragraph with the sample prepared as given in TABLE III.

TABLE III. Standard samples

	Type I and II
RDX[a]	0.5400 ± 0.005g
HMX[a]	0.0550 ± 0.005g
TNT	0.3950 ± 0.005g
Desensitizing Wax	0.0100 ± 0.001g

[a] Pure (99.8% min) RDX and Pure HMX

NOTE: If results obtained with the standard vary significantly from the actual composition, the necessary corrective actions shall be taken and the method shall not be used until it gives the proper results with the standard sample.

4.6.1.5 Determination. Inject 10 microliters of the solution into a High Performance Liquid Chromatographic Apparatus equipped with a 15.24cm (6 inch) by 4mm ID column packed with 5 micron RP-8 Li Chrosorb or equal, and a variable wavelength ultraviolet detector set at 300 nm.

Maintain a mobile phase of methanol/water/acetonitrile (49/50/1) at a flow rate of 1.0 ml/minute. Use UV grade, filtered methanol and acetonitrile and deionized, distilled and filtered water. Determine the RDX, HMX and TNT content by use of an integrator system, attached to the HPLC.

4.6.1.6 Calculation. Calculate the concentrations as follows:

$$\% \text{ TNT} = \frac{A\ (100-B)}{100} \qquad \text{(See 6.5)}$$

$$\% \text{ RDX and HMX} = \frac{C\ (100-B)}{100} \qquad \text{(See 6.5)}$$

$$\% \text{ HMX} = \frac{D\ (100-B)}{100}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:21Z
Check the source to verify that this is the current version before use.

Where:

A = The percentage of TNT from the integrator printout.
B = The percentage of wax as determined in 4.6.3.
C = The combined percentages of RDX and HMX from the integrator printout.
D = The percentage of HMX from the integrator printout.

4.6.2 RDX and HMX content (alternate method) for types I and II Comp B. Transfer a 5 to 10 gram portion of the ground sample accurately weighed to nearest 0.0001g, to a 100 ml beaker containing 30 ml of toluene saturated with RDX. Break up any lumps with a plastic coated metal rod. Cover the beaker with a watch glass and place beaker on a steam bath and heat. Stir the contents of the beaker with the glass rod. Remove the beaker and contents from the steam bath when all the solubles have dissolved (about 30 minutes) and allow beaker and contents to cool to room temperature. Quantitatively filter the solution through a tared coarse porosity filtering crucible. Use additional toluene saturated with RDX to aid in the quantitative transfer of the RDX. Use four (4) additional portions of approximately 5 ml of the saturated toluene to wash the crucible and contents. Draw air through the crucible until there is no longer a detectable odor of toluene. Dry the crucible and contents at 100°C ± 5°C for one (1) hour, cool in a desiccator and weigh. Calculate the percentage of RDX as follows:

$$\text{Percent RDX} = \frac{100\ A}{W-MW}$$

Where:

A = increase of weight of filtering crucible, g.
W = weight of sample, g.
M = percent moisture in material, expressed as a decimal (see 4.6.5)

4.6.3 Desensitizing wax. Transfer an approximately 5g portion accurately weighed to the nearest 0.0001g of the Composition B broken into small pieces of maximum dimension of 1/4" to a 500 ml beaker. Add 250 ml of acetone chilled to 5°C, to the beaker and place on a magnetic stirrer. Place a teflon stirring bar in the beaker and stir for about 10 minutes or until all of the RDX has dissolved. Before the acetone has become warm, vacuum filter the contents of the beaker through a medium porosity filtering crucible, which has been tared to the nearest 0.0001g. Quantitatively transfer any residue in the beaker to the filtering crucible with the aid of a stream of 5°C acetone. Rinse down the sides of the crucible and the residue twice with the 5°C

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:21Z
Check the source to verify that this is the current version before use.

acetone to remove any traces of RDX and aspirate the sample until dry (2 to 15 minutes). Place the crucible on a steam heated hot plate, or a vented oven maintained at 60-70°C for about five minutes. Place the crucible in a desiccator and allow to cool. Weigh the crucible and contents to the nearest 0.0001g. Calculate the percentage wax as follows:

$$\text{Wax } \% = \frac{100\ (A-B) + F}{W-MW}$$

Where:

A = the weight of the crucible and wax in g,
B = the tare weight of the crucible in g,
W = the weight of the sample in g,
M = the percent moisture in the sample, as determined in 4.6.5, expressed as a decimal, and
F = the compensating factor for the solutility of the wax in acetone as determined in 4.6.3.1.

4.6.3.1 Compensating factor for acetone solubility of wax. Prepare an approximately 0.5g sample of desensitizing wax used in the Composition B by scraping a fresh surface of a sample of the wax with a sharp knife or razor blade. Accurately weigh the sample to the nearest 0.0001g and place in a 500 ml beaker. Perform the procedure given in 4.6.3 and calculate the wax solubility factor as follows:

$$F = \frac{C-D}{W}$$

Where:

C = the weight of the crucible and wax in g before extraction,
D = the weight of the crucible and wax in g after extraction, and
W = the weight of the wax sample in g.

This procedure shall be run in triplicate and the average of the determinations shall be recorded as the wax solubility factor.

A new wax solubility factor shall be determined whenever there is a change in the lot or supplier of the wax.

4.6.4 TNT (alternate method). Determine the TNT content by difference by subtracting the sum of the percentages of RDX and HMX obtained in 4.6.1 or 4.6.2 and desensitizer content obtained in 4.6.3 from 100.0 percent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:21Z
Check the source to verify that this is the current version before use.

4.6.5 Determination of moisture content. (See 6.2.1)

4.6.5.1 Special solvent. The special solvent shall consist of equal volumes of anhydrous methanol and toluene thoroughly mixed. If necessary the solvents shall be dried by distillation.

4.6.5.2 Procedure (Karl Fischer method). Determine the moisture content in accordance with Method 101.4.1 of MIL-STD-650 using the special solvent cited in 4.6.5.1.

4.6.6 Bulk density. Determine the bulk density of the granular Composition B in accordance with Method 201.1 of MIL-STD-650.

4.6.7 Insoluble particles. Determine the insoluble particles in accordance with Method 106.1 of MIL-STD-650. Sufficient solvent shall be added to completely dissolve a 50 gram sample. Solvent should be filtered immediately prior to use.

4.6.8 Determination of viscosity (efflux method). The viscosity shall be determined in accordance with Method 212.1 of MIL-STD-650.

4.6.9 Granulation. Determine the granulation in accordance with Method 204.1 of MIL-STD-650.

4.6.10 Workmanship. Visually examine the production of the Comp B and all samples submitted for testing for compliance with the requirements given in 3.5.

5. PACKAGING

5.1 Packing.

5.1.1 Level A. Composition B shall be packed in accordance with Drawing 7548644.

5.1.2 Level B. Composition B shall be packed with Level A packing requirements.

5.1.3 Level C. Composition B shall be packed in accordance with Drawing 7548645 or 9257923.

5.2 Marking.

5.2.1 Level A. Composition B shall be marked in accordance with Drawing 7548644.

5.2.2 Level B. Composition B shall be marked in accordance with Level A marking requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:21Z
Check the source to verify that this is the current version before use.

5.2.3 Level C. Composition B shall be packed in accordance with Drawing 7548645 or 9257923.

6. NOTES.

6.1 Ordering data.

6.1.1 Acquisition requirements.

a. Title, number and date of this detail specification.
b. Quantity required and delivery schedules.
c. Type and grade required (see 1.2).
d. First article requirements - Sample provisions (see 3.3 & 4.4).
e. Level of packing protection required.

6.1.2 Data requirements. When this specification is used in a procurement contract which incorporates DD Form 1423, acceptance and description sheets prepared in accordance with MIL-STD-1171, DRSAR-P-702-109 or as directed in the contract shall be supplied by the contractor. In addition, the Process Control Document (see 3.4 and 4.2.) shall be submitted in accordance with DI-P-10392.

6.2 Equivalent test methods. The test methods given in this specification are the official methods to be used. The contractor may request using other methods providing that the proposed method is equivalent (accuracy and precision) to the method given in this specification. Prior approval of the contracting officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the contracting officer to: Commander, AMCCOM, ATTN: AMSMC-QAR-R (D), Dover, NJ 07801-5001. This description should include, but not be limited to, the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required (see MIL-I-45208).

6.2.1 Approved equivalent test methods.

6.2.1.1 Moisture determination. Holston Defense Corporation Analytical Standard Method No. C-3, dated 4 Oct 71, is approved as an equivalent test method and may be used in place of the Karl Fischer Method (4.6.5.2).

6.3 Screening for specific defects. At the direction of the contracting officer, and with the recommendation of the preparing activity (AMSMC-QAR-R(D)), batches may be screened for specific defects.

6.4 Submission of inspection equipment designs for approval. Submit designs as required to Commander, AMCCOM, ATTN: AMSMC-QAR-R (D), Dover, New Jersey, 07801-5001.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:21Z
Check the source to verify that this is the current version before use.

6.5 Correction applied to the HPLC compositional analysis. The RDX and TNT content requirements in this specification were established based on the results of the solvent extraction method (4.6.2 and 4.6.4). Since the results of the HPLC method (4.6.1) differ from, those obtained by the extraction method, a "correction" factor must be applied to them in order to bring them into line with those obtained by the extraction method.

6.6 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the custodianship of ARDEC.

6.7 Change from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army-AR

Preparing activity:
Army-AR

(Project 1376-A297)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T07:21Z
Check the source to verify that this is the current version before use.